

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1153

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing
Techniques (BL4, BL5)
Assessment Tool Weightage: 25%

QUESTIONS: 05

Total Marks:25

Annexure A

Error Analysis Task

Code of Conduct:

I K Madan Mohan Reddy(192411206) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Kyreddy Madan

RUBRICS FOR CALCULATION:

Error Analysis Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Error Identification	20%	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30%	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25%	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15%	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10%	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Error Analysis Task

Q2. Scenario: Food Delivery Application

Features:

Browse Menu, Add to Cart, Checkout

Question:

Design pseudocode for **usability testing evaluation** that measures user interaction efficiency (e.g., number of clicks, time taken to place an order). Explain how the algorithm captures and evaluates user behavior.

Usability Testing Evaluation for Food Delivery Application

Scenario

A food delivery application provides the following features:

- Browse Menu
- Add to Cart
- Checkout

The goal is to evaluate how efficiently users interact with the application by measuring:

1. Number of clicks
2. Time taken to place an order
3. Task completion success

Pseudocode

Algorithm UsabilityTestingEvaluation

Input: User performs food ordering task

Output: Usability Evaluation Report

Begin

clicks $\leftarrow$ 0

startTime $\leftarrow$ CurrentTime()

Display "Start Ordering Process"

While orderPlaced = False do

 action $\leftarrow$ GetUserAction()

 If action = Click then

 clicks $\leftarrow$ clicks + 1

 End If

 If action = BrowseMenu then

 Record("Menu Browsed")

 End If

 If action = AddToCart then

 Record("Item Added")

 End If

 If action = Checkout then

 Record("Checkout Started")

 End If

 If action = ConfirmOrder then

 orderPlaced $\leftarrow$ True

 End If

```

End While

endTime ← CurrentTime()

timeTaken ← endTime - startTime

If orderPlaced = True then
    status ← "Success"
Else
    status ← "Failure"
End If

efficiencyScore ← 100 / (clicks + timeTaken)

Print "Evaluation Report"

Print "Total Clicks =", clicks

Print "Time Taken =", timeTaken

Print "Task Status =", status

Print "Efficiency Score =", efficiencyScore

End

```

Explanation of the Algorithm

1. Initialization

The algorithm begins by setting:

- Click counter to zero.
- Start time when the user starts ordering food.

```

clicks ← 0
startTime ← CurrentTime ()

```

This allows measurement of user effort and task duration.

2. Capturing User Interactions

Every action performed by the user is monitored.

Examples:

- Browsing menu
- Selecting food items
- Adding items to cart
- Proceeding to checkout
- Confirming order

Each click increases the click counter.

Formula:

$$\textit{Time Taken} = \textit{End Time} - \textit{Start Time}$$

A lower completion time indicates better usability.

3. Measuring Task Completion Time

The algorithm measures the total time required to complete the ordering process.

The timer starts when the user begins browsing the menu and stops when the order is successfully placed.

The formula used is:

$$\textit{Time Taken} = \textit{End Time} - \textit{Start Time}$$

A shorter completion time generally indicates better usability because users can achieve their goals more quickly.

4. Tracking Task Success

An important aspect of usability testing is determining whether users can successfully complete the intended task.

The algorithm checks:

- Whether the user reached the checkout page.
- Whether payment information was entered correctly.
- Whether the order confirmation screen was displayed.

If all steps are completed successfully, the task status is marked as Success. Otherwise, it is marked as Failure.

This metric helps identify obstacles that prevent users from completing their orders.

5. Calculating User Interaction Efficiency

The efficiency score combines the number of clicks and the time taken into a single performance measure.

Formula:

Efficiency Score = $100 / (\text{Clicks} + \text{Time Taken})$

The score increases when:

- Fewer clicks are required.
- Less time is needed to place an order.

A high efficiency score indicates a user-friendly and well-designed application.

Metrics Used in Evaluation

The usability evaluation uses several important metrics:

a) Number of Clicks

Measures the amount of interaction required to complete a task.

Purpose:

- Identify unnecessary navigation steps.
- Evaluate interface simplicity.

b) Time Taken

Measures how quickly users complete the ordering process.

Purpose:

- Determine system efficiency.
- Compare performance among different users.

c) Task Completion Rate

Measures whether users successfully place an order.

Formula:

Task Completion Rate =

$(\text{Number of Successful Orders} / \text{Total Attempts}) \times 100$

Purpose:

- Assess effectiveness of the application.

d) Efficiency Score

Combines click count and completion time to evaluate overall performance.

Example Scenario

Consider a user performing the following actions:

- Browse Menu
- Select Pizza
- Add Pizza to Cart
- Open Cart
- Proceed to Checkout
- Enter Address

- Select Payment Method

How the Algorithm Captures User Behavior

- The proposed algorithm captures user behavior in the following ways:
- Records every user interaction.
- Tracks navigation patterns across application screens.
- Measures the time spent on each task.
- Identifies successful and unsuccessful task completions.
- Calculates efficiency metrics for performance analysis.
- Helps detect usability issues such as excessive clicks or long completion times.
- Provides quantitative data for improving application design.
- Confirm Order

Advantages of the Evaluation Method

- Simple and easy to implement.
- Provides measurable usability metrics.
- Helps improve user satisfaction.
- Identifies inefficient navigation paths.
- Supports data-driven interface improvements.
- Enhances overall application performance.

Conclusion

Usability testing is essential for evaluating the effectiveness and efficiency of a food delivery application. The proposed algorithm measures user interaction efficiency by recording clicks, tracking completion time, and verifying successful task completion. By analyzing these metrics, developers can identify usability problems and optimize the application's interface, resulting in a faster, easier, and more satisfying user experience.

